2 November 1972
SUPERSEDING
MIL-H-1431A
13 December 1963

MILITARY SPECIFICATION

HAMMER, HAND (PITON, MOUNTAIN)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type and size of piton hammer.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-H-93	-	Handles, Hickory, Striking-Tool
TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces or Organic Coatings
TT-E-485	-	Enamel, Semi-Gloss, Rust-Inhibiting
PPP-P-40	-	Packaging and Packing of Hand Tools

MILITARY

MIL-C-5040	-	Cord, Nylon

STANDARDS

FEDERAL

FED-STD-66	-	Steel: Chemical Composition and Hardenability
FED-STD-595	-	Colors

FSC 5120

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

DRAWINGS

NATICK LABORATORIES

2-10-57 - Hammer, Piton, Mountain

(Miniature copy of Drawing 2-10-57, identified as figure 1, is attached for information purposes only.)

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials (ASTM)

E.18-68 - Methods of Test for Rockwell Hardness and Superficial Hardness of Metallic Materials

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), the supplier shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.4 and 6.3).

3.2 Materials and components. Materials shall conform to the referenced specifications, standards and requirements specified herein.

3.2.1 Head. The head shall be fabricated from steel conforming to AISI Steel Number 1060 of Standard FED-STD-66.

3.2.2 Plates. The reinforcing plates shall be fabricated from steel conforming to AISI Steel Number 1020 of Standard FED-STD-66.

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* 3.2.3 Cord. The cord may be type I, IA or IIA conforming to MIL-C-5040.

3.2.4 Handles. The handle shall conform to the wood requirements for grade A handles of Specification NN-H-93 and to the dimensions shown on Drawing 2-10-57.

3.2.5 Thread polyester. The thread shall conform to type I or II, class 1, size E of Specification V-T-285.

3.3 Construction. The construction shall be as specified herein and shall conform to the design, dimensions, and materials shown on Drawing 2-10-57.

* 3.3.1 Hammer head. The hammer head shall be an integral forging from steel. The shape shall conform to the dimensions shown on the drawing. The striking face and pick shall be hardened and tempered to a hardness of Rockwell C25 to C50. Determination of this requirement shall be as specified in 4.7.1. Case hardened heads shall not be acceptable.

* 3.3.2 Assembly. The handle shall be inserted into the hammer head between the reinforcing plates specified in 3.2.2 and secured with a metal wedge having ribbed or corrugated construction. The wedge shall be driven flush within plus or minus 1/32 inch of the top of handle. The wedge, after the reinforcing plates are in place, shall create sufficient expansion to completely fill the tool eye and tightly wedge the handle into the eye of the hammer head. The rivet shall be driven, peened, and shall hold the reinforcing plates tight against the corresponding flat surface of the handle. The hammer head shall be at a 90 degree angle with the center line of the handle. The ends of cord shall be sewn or seared together as shown on Drawing 2-10-57.

3.3.3 Stitching, machine. Stitching shall conform to FED-STD-751, type 304, 14 to 16 stitches per inch. Stitching shall be as shown on Drawing 2-10-57.

3.3.4 Type 304 stitching. Thread breaks in stitching shall be overstitched not less than 3/4 inch at each break. Thread tension shall be maintained so that there will be no loose stitching.

3.4 Finish.

3.4.1 Surface preparation.

* 3.4.1.1 Handle. Prior to assembly, the handle shall be smoothly sanded, cleaned, and given a coat of sealer or wood preservative of the quality normally used by the supplier.

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* 3.4.1.2 Metal. All metal shall be protected against rusting. Reinforcing strips shall be prepared in accordance with any method or type specified in TT-C-490 and then protected by rust-resisting coating. After heat treating, the head, if not painted, shall be prepared and protected as normally furnished by the supplier. If painted, at the option of the supplier, the heat treated head shall be prepared for paint in accordance with any method or type of TT-C-490.

3.5 Painting.

* 3.5.1 Hammer. Painted hammers shall have an enamel coat conforming to TT-E-485. The color shall be olive drab, corresponding to Color Number 24087 of FED-STD-595.

3.6 Identification marking. Each hammer shall be plainly and permanently marked in the size and location shown on the drawing.

* 3.7 Workmanship. All components of the hammer shall be free from dirt, burrs, slivers, rough die, tool and grind marks. Forgings and stampings shall be free of sand, pins, pits, blow holes and spruces. External surfaces shall be free from sharp edges and corners, except when corners are required.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Component and material inspection. The supplier shall insure that all components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards unless excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).
c. Inspection of preparation for delivery (see 4.6).

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* 4.4 First article. When a first article is required, the hammer shall be examined for defects in table I, dimensions specified, and tested as specified in 4.7. The presence of any defect and failure of any test shall be cause for rejection of the first article.

* 4.5 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

* 4.5.1 In-process inspection.

4.5.1.1 Process examination. Examination shall be made to determine that the sealer is applied to the handle (see 3.4.1.1) and preparation for finish is made to the hammer head and reinforcing strips (see 3.4.1.2). Whenever a deviation is noted, correction shall be made to the affected items and lot in process.

4.5.1.2 Intermediate testing. The hammer head shall be tested as specified in 4.7.1. The sample unit shall be one hammer head. An inspection lot shall consist of all hammer heads offered for inspection at one time. The inspection level shall be S-3 with an acceptable quality level (AQL) of 1.5 defects, expressed in terms of defects per hundred units.

* 4.5.2 End item inspection. The inspection lot shall be all hammers of the same design offered for inspection at one time.

* 4.5.2.1 Visual examination. The hammer shall be examined for defects in table I. The inspection level shall be level II with an AQL of 2.5 per major and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not as specified.	X	
	Peeling, blistering or chipping.		X
	Finish tacky, not completely dry.		X
Construction and workmanship	Sharp burr.	X	
	Handle not sanded smoothly.	X	
	Reinforcing plate missing.	X	
	Components not properly fitted (unless otherwise classified).	X	
	Wedge not flush within plus or minus 1/32 inch of handle end.	X	
	Rivet missing.	X	
	Rivet loose.		X
	Rivet not peened.	X	

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Cord	Missing.	X	
	Ends not joined (stitched or knotted).	X	
Stitch type	Wrong stitch type.	X	
Stitch tension	Loose, resulting in a loosely exposed top or bottom (bobbin or looper) thread.		X
	Tight, resulting in excessive tightness.		X
Stitches per inch	Two or more stitches less than minimum specified.		X
	Two or more stitches in excess of maximum specified.		X
Thread breaks	Overstitched less than 3/4 inch on stitch type 304.		X
Marking	Missing, incomplete, not legible.		X

4.5.2.2 Dimensional examination. Examination shall be made of the hammer to determine compliance with dimensional requirements. Any dimension not within the established requirement shall be classified a defect. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

* 4.6 Inspection of preparation for delivery. Examination shall be made to determine compliance with preservation and packaging, packing and marking requirements in accordance with PPP-P-40.

4.7 Test.

* 4.7.1 Hardness test. The hardness test shall be performed in accordance with ASTM E.18-68 to determine compliance with 3.3.1. One determination shall be made for each requirement. The impression for the pick end shall be not closer than 1/4 inch from the point. Any reading not within limits specified shall constitute failure of this test.

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5. PREPARATION FOR DELIVERY

5.1 Military requirements.

* 5.1.1 Levels A, B and C. Preservation, packaging and packing for each level of protection shall be in accordance with PPP-P-40.

6. NOTES

6.1 Intended use. The piton hammer covered by this specification is intended to be used for driving all types of ice and rock pitons covered by MIL-P-1474.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.1, 4.4 and 6.3).
(c) Selection of applicable levels of preservation and packaging and packing (see 5.1).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 Revision changes. The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Air Force - 84

User activity:

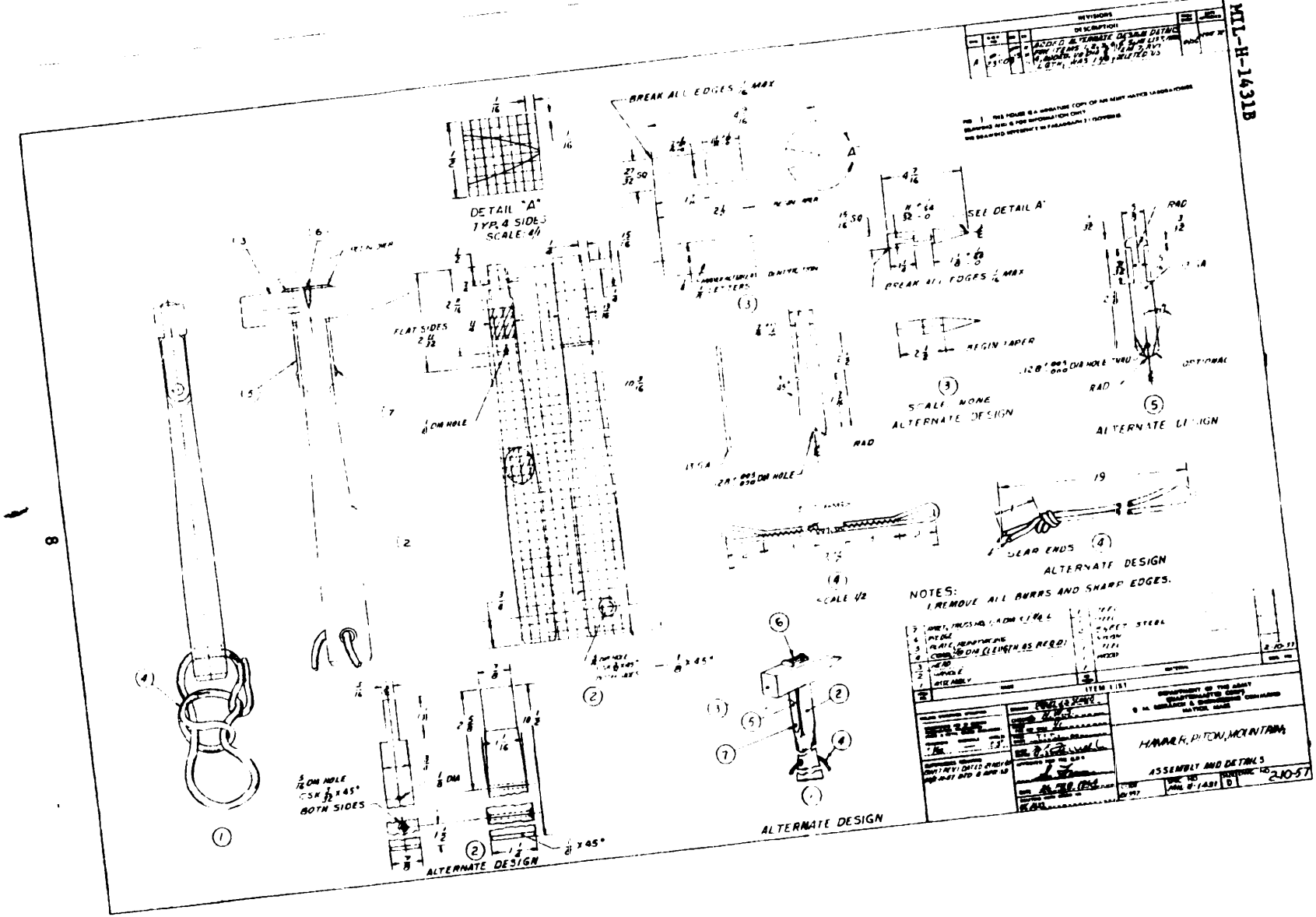
Army - MD

Preparing activity:

Army - GL

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION HAmmer, Hand (Piton, Mountain) MIL-H-1431B

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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